MB4403 WEEK 4 2020/21

UNDETERMINED COEFFICIENTS: ANNIHILATOR APPROACH

Recall that $D^n y$ is one way to denote $\frac{d^n y}{dx^n}$

Consider a linear nonhomogeneous equation with constant coefficients

$$a_n D^n y + a_{n-1} D^{n-1} y + \cdots + a_1 D y + a_0 y = f(x), \quad f(x) \neq 0 \quad \text{---} (1)$$

For convenience, we can rewrite eqn (1) as

$$L(y) = f(x),$$

where L denotes the linear n^{th} order differential operator

$$a_n D^n + a_{n-1} D^{n-1} + \cdots + a_1 D + a_0$$

We also choose to treat D as an algebraic quantity so that if possible, L can be factored.

e.g. $L(y) = D^2 y + 3Dy - 4y = (D^2 + 3D - 4)y = (D+4)(D-1)y$

ANNIHILATOR OPERATOR

If L is a linear differential operator with constant coefficients, and the function $f(x)$ is such that

$$L(f(x)) = 0,$$

then L is called the annihilator of $f(x)$

Examples

(1) $D^3(x^2 + 2x + 1) = D^2(2x + 2)$

$= D(2)$

$= 0$

$\therefore D^3$ annihilates $x^2 + 2x + 1$

and their linear combin

In general, D^n annihilates the following functions,
$C, x, x^2, x^3, \ldots, x^{n-1}$ where C is a constant
and n is a positive integer

② $(D-\alpha)^2(xe^{\alpha x}) = (D^2 - 2D\alpha + \alpha^2)(xe^{\alpha x})$

$$D(xe^{\alpha x}) = x\alpha e^{\alpha x} + e^{\alpha x}$$
$$D^2(xe^{\alpha x}) = \alpha^2 xe^{\alpha x} + 2\alpha e^{\alpha x}$$

so
$$(D-\alpha)^2(xe^{\alpha x}) = D^2(xe^{\alpha x}) - 2D(xxe^{\alpha x}) + \alpha^2 x e^{\alpha x}$$
$$= \alpha^2 xe^{\alpha x} + 2\alpha e^{\alpha x} - 2x\alpha^2 e^{\alpha x} - 2\alpha e^{\alpha x} + \alpha^2 xe^{\alpha x}$$
$$= 0$$

$\therefore (D-\alpha)^2$ annihilates $xe^{\alpha x}$

In general, the differential operator $(D-\alpha)^n$ annihilates the following functions and their linear combinations

$$e^{\alpha x}, xe^{\alpha x}, x^2 e^{\alpha x}, \ldots, x^{n-1} e^{\alpha x}$$

3. $[D^2 - 2\alpha D + (\alpha^2+\beta^2)]\, e^{\alpha x}\cos\beta x$

$$D(e^{\alpha x}\cos\beta x) = \alpha e^{\alpha x}\cos\beta x - \beta e^{\alpha x}\sin\beta x$$
$$D^2(e^{\alpha x}\cos\beta x) = (\alpha^2-\beta^2)e^{\alpha x}\cos\beta x - 2\alpha\beta e^{\alpha x}\sin\beta x$$

$$[D^2 - 2\alpha D + (\alpha^2+\beta^2)]\, e^{\alpha x}\cos\beta x$$
$$= (\alpha^2-\beta^2)e^{\alpha x}\cos\beta x - 2\alpha\beta e^{\alpha x}\sin\beta x - 2\alpha^2 e^{\alpha x}\cos\beta x$$
$$+ 2\alpha\beta e^{\alpha x}\sin\beta x + (\alpha^2+\beta^2)e^{\alpha x}\cos\beta x$$
$$= 0$$

so $[D^2 - 2\alpha D + (\alpha^2+\beta^2)]$ is the annihilator of
$e^{\alpha x}\cos\beta x$

In general, the differential operator $[D^2 - 2\alpha D + (\alpha^2+\beta^2)]^n$ annihilates the following functions and their linear combinations

$$e^{\alpha x}\cos\beta x;\ xe^{\alpha x}\cos\beta x, x^2e^{\alpha x}\cos\beta x, \ldots, x^{n-1}e^{\alpha x}\cos\beta x$$
$$e^{\alpha x}\sin\beta x,\ xe^{\alpha x}\sin\beta x,\ x^2e^{\alpha x}\sin\beta x, \ldots, x^{n-1}e^{\alpha x}\sin\beta x$$

If L_1 annihilates y_1 and L_2 annihilates y_2, but $L_1(y_2)\neq 0$ and $L_2(y_1)\neq 0$ i.e L_1 and L_2 don't annihilate y_2 and y_1 respectively, then the product L_1L_2 annihilates the linear combination of y_1 and y_2 i.e $c_1y_1+c_2y_2$.

This is so because

$$\begin{aligned} L_1L_2(c_1y_1+c_2y_2) &= c_1L_2L_1(y_1) + c_2L_1L_2(y_2) \\ &= c_1L_2(0) + c_2L_1(0) \\ &= 0 \end{aligned}$$

Previously, we looked at the right hand side of the equation and considered the sole of the auxiliary to determine the form of the particular solution. In this approach, we first find the operator that annihilates the right hand side, and then determine the form of y_p.

ANNIHILATOR APPROACH SUMMARISED

Consider differential equation

$$L(y) = f(x) \quad \text{---} \quad (1)$$

where L is a linear differential operator with constant coefficients and $f(x)$ is linear combination of polynomials, exponentials,

sines and cosines, and their products.

Procedure:

1. Find the complementary solution y_c of eqn ①
   i.e solve $L(y)=0$
2. Find the annihilator L_1 of $f(x)$ and apply it on both sides of eqn ① to get a homogeneous equation $L_1 L(y)=0$
3. Find the solution to the homogeneous equation $L_1 L(y)=0$. This soln is the general soln of $L(y)=g(x)$. Some terms from this solution will be duplicates of terms of the solution in step 1, y_p is equal to all the remaining terms that are not duplicates.
4. Substitute y_p into the equation $L(y)=f(x)$ to determine the unknown coefficients

Examples

Solve

① $y'' - 9y = 54$

② $y'' + 4y' + 4y = 2x+6$

3. $y'' - y' - 12y = e^{4x}$

4 $y'' + 25y = 6\sin x$

Soln

①. The equation is

$$(D^2 - 9)y = 54$$

The auxiliary equation is

$$m^2 - 9 = 0 \Rightarrow m = \pm 3$$

$\therefore y_c = c_1 e^{-3x} + c_2 e^{3x}$

The annihilator of 54 is D, so

$$D(D^2-9)y = D(54)$$
$$\Rightarrow D(D^2-9)y = 0$$

The auxiliary eqⁿ is

$$m(m^2-9) = 0$$
$$\Rightarrow m = 0 \text{ or } m = \pm 3$$

∴ The general solution is

$$y = \underbrace{C_3 + C_1 e^{-3x} + C_2 e^{3x}}_{y_c}$$

So $y_p = C_3$

$$\Rightarrow y_p' = 0$$
$$\Rightarrow y_p'' = 0$$

Substituting, we get

$$-9C_3 = 54$$
$$\Rightarrow C_3 = -6$$

Thus the general solution is

$$y = C_1 e^{-3x} + C_2 e^{3x} - 6$$

(2). $(D^2+4D+4)y = 2x+6$

The auxiliary equation:

$$m^2+4m+4 = 0$$
$$\Rightarrow (m+2)^2 = 0$$
$$\Rightarrow m = -2 \quad \text{(multiplicity 2)}$$

∴ $y_c = C_1 e^{-2x} + C_2 x e^{-2x}$

$$D^2(D^2+4D+4)y = 0$$

The auxiliary eqⁿ:

$$m^2(m^2+4m+4) = 0$$

$\Rightarrow m = 0$ (multiplicity 2) or $m = -2$ (multiplicity 2)

The general solution is

$$y = c_3 + c_4 x + c_1 e^{-2x} + c_2 x e^{-2x}$$

Thus

$y_p = c_3 + c_4 x$

$\Rightarrow y_p' = c_4$

$y_p'' = 0$

Substituting

$4c_4 + 4c_3 + 4c_4 x = 2x + 6$

$4c_4 = 2 \Rightarrow c_4 = \frac{1}{2}$

$4c_4 + 4c_3 = 6$

$\Rightarrow c_3 = 1$

$\therefore y = 1 + \frac{1}{2}x + c_1 e^{-2x} + c_2 x e^{-2x}$

3. $(D^2 - D - 12)y = e^{4x}$

Auxiliary eqn: $m^2 - m - 12 = 0$

$\Rightarrow (m-4)(m+3) = 0$

$\Rightarrow m = 4$ or $m = -3$

$\therefore y_c = c_1 e^{-3x} + c_2 e^{4x}$

$(D-4)(D^2 - D - 12)y = 0$

Auxiliary eqn:

$(m-4)(m^2 - m - 12) = 0$

$m = 4$ (multiplicity 2) or $m = -3$

$\therefore y = c_2 e^{-3x} + c_1 e^{4x} + c_3 x e^{4x}$

Thus $y_p = c_3 x e^{4x}$

$\Rightarrow y_p' = c_3 e^{4x} + 4c_3 x e^{4x}$

$\Rightarrow y_p'' = 4C_3e^{4x} + 4C_3e^{4x} + 16C_3xe^{4x}$

$= 8C_3e^{4x} + 16C_3xe^{4x}$

Substituting

$8C_3e^{4x} + 16C_3xe^{4x} - C_3e^{4x} - 4C_3xe^{4x} - 12C_3xe^{4x} = e^{4x}$

$\Rightarrow 7C_3e^{4x} = e^{4x}$

$\Rightarrow 7C_3 = 1$

$\Rightarrow C_3 = \frac{1}{7}$

$\therefore y = C_1e^{4x} + C_2e^{-3x} + \frac{1}{7}xe^{4x}$

4. $(D^2+25)y = 6\sin x$

Auxiliary eqⁿ:

$m^2+25=0 \Rightarrow m = \pm 5i$

$\therefore y_c = C_1\cos 5x + C_2\sin 5x$

$[D^2+1](D^2+25)y = 0$

Auxiliary eqⁿ:

$(m^2+1)(m^2+25) = 0$

$m = \pm i$ or $m = \pm 5$

$\therefore y = C_3\cos x + C_4\sin x + C_1\cos 5x + C_2\sin 5x$

Thus $y_p = C_3\cos x + C_4\sin x$

$\Rightarrow y_p' = -C_3\sin x + C_4\cos x$

$\Rightarrow y_p'' = -C_3\cos x - C_4\sin x$

Substituting

$-C_3\cos x - C_4\sin x + 25C_3\cos x + 25C_4\sin x = 6\sin x$

$\Rightarrow 24C_3\cos x + 24C_4\sin x = 6\sin x$

$\Rightarrow C_3 = 0$ and $C_4 = \frac{1}{4}$

$\therefore y = C_1 \cos 5x + C_2 \sin 5x + \frac{1}{4}\sin x$

2. VARIATION OF PARAMETERS

Unlike the undetermined coefficients method where the right hand side is expected to be certain limited types of function and the equation should have constant coefficients, variation of parameters can always be used to find a particular solution provided that the associated homogeneous equation can be solved.

PROCEDURE:

Lets consider a second order nonhomogeneous eq^{n}

$$a y'' + b y' + c y = f(x) \quad \text{—} \quad (1)$$

To find a particular solution to this eq^{n}

(1). Find the complementary solution $y_c = C_1 y_1 + C_2 y_2$

(2). Divide (1) by a to get the form

$$y'' + By' + Dy = g(x) \quad \text{where } B = \frac{b}{a},\ D = \frac{c}{a},\ g(x) = \frac{f(x)}{a}$$

(3). Compute the Wronskian $W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$ and expressions W_1 and W_2, where

$$W_1 = \begin{vmatrix} 0 & y_2 \\ g(x) & y_2' \end{vmatrix}, \quad W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & g(x) \end{vmatrix}$$

(4). Integrate $u_1' = \frac{W_1}{W}$ and $u_2' = \frac{W_2}{W}$ to get u_1 and u_2. Constants of integration can be taken as zero since we are looking for at least one

solution, so one is enough. The particular solution is given by $y_p = u_1 y_1 + u_2 y_2$, and the general solution is $y = y_c + y_p$.

For higher-order differential equations

$$u_k' = \frac{W_k}{W}, \quad k = 1, 2, \cdots n$$

$$W = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & & y_n' \\ y_1'' & y_2'' & & y_n'' \\ \vdots & \vdots & & \vdots \\ y_1^{n-1} & y_2^{n-1} & & y_n^{n-1} \end{vmatrix}, \quad W_1 = \begin{vmatrix} 0 & y_2 & y_3 & \cdots & y_n \\ 0 & y_2' & y_3' & & y_n' \\ \vdots & y_2'' & y_3'' & & y_n'' \\ 0 & \vdots & \vdots & & \vdots \\ f(x) & y_2^{n-1} & y_3^{n-1} & & y_n^{n-1} \end{vmatrix}$$

$$W_2 = \begin{vmatrix} y_1 & 0 & y_3 & \cdots & y_n \\ y_1' & 0 & y_3' & & y_n' \\ y_1'' & \vdots & y_3'' & & y_n'' \\ \vdots & 0 & \vdots & & \vdots \\ y_1^{n-1} & f(x) & y_3^{n-1} & & y_n^{n-1} \end{vmatrix}, \quad \cdots \quad W_k = \begin{vmatrix} y_1 & y_2 & \cdots & 0 & y_{k+1} & \cdots & y_n \\ y_1' & y_2' & & \vdots & y_{k+1}' & & y_n' \\ \vdots & \vdots & & 0 & \vdots & & \vdots \\ y_1^{n-1} & y_2^{n-1} & & f(x) & y_{k+1}^{n-1} & & y_n^{n-1} \end{vmatrix}$$

$$W_n = \begin{vmatrix} y_1 & y_2 & \cdots & y_{n-1} & 0 \\ y_1' & y_2' & & y_{n-1}' & \vdots \\ \vdots & \vdots & & \vdots & 0 \\ y_1^{n-1} & y_2^{n-1} & & y_{n-1}^{n-1} & f(x) \end{vmatrix}$$

Examples

Find the general solution

① $y'' + y = \sec x$

② $y'' + 2y' + y = e^{-x}$

③ $3y'' - 6y' + 6y = e^{x}\sec x$

④ $y''' - 2y'' - y' + 2y = e^{4x}$

⑤ $y''' + y' = \tan x$

Soln

① The auxiary equation is

$m^2 + 1 = 0$

$\Rightarrow m = \pm i$

$y_c = c_1 \cos x + c_2 \sin x$

$$W(\cos x, \sin x) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1$$

$$W_1 = \begin{vmatrix} 0 & \sin x \\ \sec x & \cos x \end{vmatrix} = -\tan x$$

$$W_2 = \begin{vmatrix} \cos x & 0 \\ -\sin x & \sec x \end{vmatrix} = 1$$

$u_1' = -\tan x$ and $u_2' = 1$

$\Rightarrow u_1 = \ln(\cos x)$ $\Rightarrow u_2 = x$

So $y_p = \ln(\cos x)\cos x + x \sin x$

$\therefore y = c_1 \cos x + c_2 \sin x + \cos x \ln \cos x + x \sin x$

(2) The auxiliary eq^n –

$$m^2 + 2m + 1 = 0$$

$$\Rightarrow (m+1)^2 = 0$$

$$\Rightarrow m = -1 \text{ (multiplicity 2)}$$

$$\therefore y_c = c_1 e^{-x} + c_2 x e^{-x}$$

$$W(e^{-x}, xe^{-x}) = \begin{vmatrix} e^{-x} & xe^{-x} \\ -e^{-x} & e^{-x} - xe^{-x} \end{vmatrix}$$

$$= e^{-2x} - xe^{-2x} + xe^{-2x}$$

$$= e^{-2x}$$

$$W_1 = \begin{vmatrix} 0 & xe^{-x} \\ e^{-x} & e^{-x} - xe^{-x} \end{vmatrix}$$

$$= -xe^{-2x}$$

$$W_2 = \begin{vmatrix} e^{-x} & 0 \\ -e^{-x} & e^{-x} \end{vmatrix}$$

$$= e^{-2x}$$

$$u_1' = \frac{-xe^{-2x}}{e^{-2x}}, \quad u_2' = \frac{e^{-2x}}{e^{-2x}}$$

$$\Rightarrow u_1' = -x, \quad u_2' = 1$$

$$\Rightarrow u_1 = -\frac{1}{2}x^2, \quad u_2 = x$$

$$\Rightarrow y_p = -\frac{1}{2}x^2 e^{-x} + x^2 e^{-x} = \frac{1}{2}x^2 e^{-x}$$

$$\therefore y = c_1 e^{-x} + c_2 x e^{-x} + \frac{1}{2}x^2 e^{-x}$$

③ · The auxiliary eq^{n}:

$$3m^2 - 6m + 6 = 0$$

$$m = \frac{6 \pm \sqrt{36 - 4(3)6}}{6}$$

$$= 1 \pm i$$

$$y_c = C_1 e^x \cos x + C_2 e^x \sin x$$

Rewriting the equation, we have

$$y'' - 2y' + 2y = \frac{1}{3} e^x \sec x$$

$$W = \begin{vmatrix} e^x \cos x & e^x \sin x \\ e^x \cos x - e^x \sin x & e^x \sin x + e^x \cos x \end{vmatrix}$$

$$= e^{2x}\left[\cos x \sin x + \cos^2 x\right] - e^{2x}\left[\cos x \sin x - \sin^2 x\right]$$

$$= e^{2x}$$

$$W_1 = \begin{vmatrix} 0 & e^x \sin x \\ \frac{1}{3} e^x \sec x & e^x \sin x + e^x \cos x \end{vmatrix}$$

$$= -\frac{1}{3} e^{2x} \tan x$$

$$W_2 = \begin{vmatrix} e^x \cos x & 0 \\ e^x \cos x - e^x \sin x & \frac{1}{3} e^x \sec x \end{vmatrix}$$

$$= \frac{1}{3} e^{2x}$$

$$u_1' = -\frac{1}{3} \tan x \quad \text{and} \quad u_2' = \frac{1}{3}$$

$$\Rightarrow u_1 = \frac{1}{3} \ln(\cos x) \quad \text{and} \quad u_2 = \frac{1}{3} x$$

$$y_p = \frac{1}{3}\ln(\cos x)\, e^x \cos x + \frac{1}{3} x e^x \sin x$$

$$\therefore y = C_1 e^x \cos x + C_2 e^x \sin x + \frac{1}{3} e^x \cos(x) \ln(\cos x) + \frac{1}{3} x e^x \sin x$$

(H). The auxiliary eqⁿ

$$m^3 - 2m^2 - m + 2 = 0$$

$$\Rightarrow (m-1)(m^2 - m - 2) = 0$$

$$\Rightarrow (m-1)(m-2)(m+1) = 0$$

$$\Rightarrow m = 1, \quad m = 2 \text{ or } m = -1$$

$$\therefore y_c = C_1 e^x + C_2 e^{2x} + C_3 e^{-x}$$

$$W(e^x, e^{2x}, e^{-x}) = \begin{vmatrix} e^x & e^{2x} & e^{-x} \\ e^x & 2e^{2x} & -e^{-x} \\ e^x & 4e^{2x} & e^{-x} \end{vmatrix}$$

$$= e^x \begin{vmatrix} 2e^{2x} & -e^{-x} \\ 4e^{2x} & e^{-x} \end{vmatrix} - e^x \begin{vmatrix} e^{2x} & e^{-x} \\ 4e^{2x} & e^{-x} \end{vmatrix} + e^x \begin{vmatrix} e^{2x} & e^{-x} \\ 2e^{2x} & -e^{-x} \end{vmatrix}$$

$$= e^x\left[2e^x + 4e^x\right] - e^x\left[e^x - 4e^x\right] + e^x\left[-e^x - 2e^x\right]$$

$$= 6e^{2x} + 3e^{2x} - 3e^{2x}$$

$$= 6e^{2x}$$

$$W_1 = \begin{vmatrix} 0 & e^{2x} & e^{-x} \\ 0 & 2e^{2x} & -e^{-x} \\ e^{4x} & 4e^{2x} & e^{-x} \end{vmatrix}$$

$$= e^{4x} \begin{vmatrix} e^{2x} & e^{-x} \\ 2e^{2x} & -e^{-x} \end{vmatrix}$$

$$= e^{4x}[-e^{x} - 2e^{x}]$$

$$= e^{4x}(-3e^{x})$$

$$= -3e^{5x}$$

$$W_2 = \begin{vmatrix} e^{x} & 0 & e^{-x} \\ e^{x} & 0 & -e^{-x} \\ e^{x} & e^{4x} & e^{-x} \end{vmatrix}$$

$$= e^{4x} \begin{vmatrix} e^{x} & e^{-x} \\ e^{x} & -e^{-x} \end{vmatrix}$$

$$= e^{4x}[-1 - 1]$$

$$= -2e^{4x}$$

$$W_3 = \begin{vmatrix} e^{x} & e^{2x} & 0 \\ e^{x} & 2e^{2x} & 0 \\ e^{x} & 4e^{2x} & e^{4x} \end{vmatrix}$$

$$= e^{4x} \begin{vmatrix} e^{x} & e^{2x} \\ e^{x} & 2e^{2x} \end{vmatrix}$$

$$= e^{4x}[2e^{3x} - e^{3x}]$$

$$= e^{7x}$$

$$u_1' = -\frac{1}{2}e^{3x}, \quad u_2' = -\frac{1}{3}e^{2x}, \quad u_3' = \frac{e^{5x}}{6}$$

$$\Rightarrow u_1 = \frac{1}{6}e^{3x}, \quad u_2 = -\frac{1}{6}e^{2x}, \quad u_3 = \frac{1}{30}e^{5x}$$

$$y_p = \frac{1}{6}e^{4x} - \frac{1}{6}e^{4x} + \frac{1}{30}e^{4x} = \frac{1}{30}e^{4x}$$

$$\therefore y = C_1 e^{x} + C_2 e^{2x} + C_3 e^{-x} + \frac{1}{30}e^{4x}$$

(5). Auxiliary eqn

$$m^3 + m = 0$$

$$\Rightarrow m(m^2+1) = 0$$

$$\Rightarrow m = 0 \text{ or } m = \pm i$$

$$\therefore y_c = C_1 + C_2\cos x + C_3 \sin x$$

$$W(1, \cos x, \sin x) = \begin{vmatrix} 1 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ 0 & -\cos x & -\sin x \end{vmatrix}$$

$$= \begin{vmatrix} -\sin x & \cos x \\ -\cos x & -\sin x \end{vmatrix}$$

$$= \sin^2 x + \cos^2 x$$

$$= 1$$

$$W_1 = \begin{vmatrix} 0 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ \tan x & -\sin x & -\sin x \end{vmatrix}$$

$$= \tan x \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$$

$$= \tan x$$

$$W_2 = \begin{vmatrix} 1 & 0 & \sin x \\ 0 & 0 & \cos x \\ 0 & \tan x & -\sin x \end{vmatrix}$$

$$= \begin{vmatrix} 0 & \cos x \\ \tan x & -\sin x \end{vmatrix}$$

$$= -\sin x$$

$$W_3 = \begin{vmatrix} 1 & \cos x & 0 \\ 0 & -\sin x & 0 \\ 0 & -\cos x & \tan x \end{vmatrix}$$

$$= \begin{vmatrix} -\sin x & 0 \\ -\cos x & \tan x \end{vmatrix}$$

$$= -\sin(x)\tan x$$

$u_1' = \tan x$, $u_2' = -\sin x$, $u_3' = -\sin x \tan x$

$\Rightarrow u_1 = -\ln(\cos x)$, $u_2 = \cos x$, $u_3' = -\dfrac{\sin^2 x}{\cos x}$

$$= -\frac{(1 - \cos^2 x)}{\cos x}$$

$$= -\sec x + \cos x$$

$$\Rightarrow u_3 = -\ln|\sec x + \tan x| + \sin x$$

$$y_p = -\ln(\cos x) + \cos^2 x + \sin^2 x - \sin x \ln|\sec x + \tan x|$$

$$= -\ln(\cos x) - \sin x \ln|\sec x + \tan x|$$

$$\therefore y = c_1 + c_2 \cos x + c_3 \sin x - \ln(\cos x) - \sin x \ln|\sec x + \tan x|$$

CAUCHY-EULER EQUATIONS (also known as equidimensional equations)

These are equations of the form

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 x \frac{dy}{dx} + a_0 y = f(x)$$

where $a_n, a_{n-1}, \ldots, a_1, a_0$ are constants

To find the complementary equation, we try the solution $y = x^m$, where m is the solution to the auxiliary equation.

Consider $a_2 x^2 y'' + a_1 x y' + a_0 y = 0$

Substituting $y = x^m$, we get

$$a_2 x^2 (m(m-1)x^{m-2}) + a_1 x (m x^{m-1}) + a_0 x^m = 0$$

$$\Rightarrow a_2 m(m-1) x^m + a_1 m x^m + a_0 x^m = 0$$

$$\Rightarrow (a_2 m(m-1) + a_1 m + a_0) x^m = 0$$

The auxiliary equation is

$$a_2 m(m-1) + a_1 m + a_0 = 0$$

REAL AND DISTINCT ROOTS OF Auxiliary Eqn

If m_1 and m_2 are roots of the auxiliary eqn and $m_1 \neq m_2$, then the solution of the homogeneous eqn is

$$y = C_1 x^{m_1} + C_2 x^{m_2}$$

REPEATED ROOTS

If m_1 and m_2 are roots ~~and~~ of the auxiliary equation and $m_1 = m_2$, then the solution is

$$y = C_1 x^{m_1} + C_2 x^{m_1} \ln x$$

COMPLEX ROOTS

If $m_1 = \alpha + i\beta$ and $m = \alpha - i\beta$ are roots of the auxiliary eqⁿ, then the solⁿ of the homogeneous equation is

$$y = C_1 x^{\alpha} \cos(\beta \ln x) + C_2 x^{\alpha} \sin(\beta \ln x)$$

Example 1

solve

(1). $x^2 y'' - 2y = 0$

(2) $x^3 y''' - 6y = 0$

Sol^n

Substituting $y = x^m$ we get

$x^2 (m(m-1)x^{m-2}) - 2x^m = 0$

The auxiliary eqⁿ is

$m(m-1) - 2 = 0$

$\Rightarrow m^2 - m - 2 = 0$

$\Rightarrow (m-2)(m+1) = 0$

$\Rightarrow m = 2$ or $m = -1$

$\therefore y = C_1 x^2 + C_2 x^{-1}$

2. Substituting $y = x^m$ we get

$x^3 m(m-1)(m-2) x^{m-3} - 6x^m = 0$

The auxiliary eqⁿ is

$m(m-1)(m-2) - 6 = 0$

$\Rightarrow m^3 - 3m^2 + 2m - 6 = 0$

$\Rightarrow (m-3)(m^2+2) = 0$

$\Rightarrow m = 3$ or $m = \pm i\sqrt{2}$

$\therefore y = C_1 x^3 + C_2 \cos(\sqrt{2}\ln x) + C_3 \sin(\sqrt{2}\ln x)$

Example 2

Use variation of parameters to solve

① $x^2y'' - xy' + y = 2x$

② $x^2y'' + xy' - y = \ln x$

Soln

① $x^2y'' - xy' + y = 0$

Substituting $y = x^m$

$x^2 m(m-1) x^{m-2} - x m x^{m-1} + x^m = 0$

The auxiliary eqⁿ is

$m(m-1) - m + 1 = 0$

$\Rightarrow m^2 - m - m + 1 = 0$

$\Rightarrow m^2 - 2m + 1 = 0$

$\Rightarrow (m-1)^2 = 0$

$\Rightarrow m = 1$ (multiplicity 2)

$\therefore y_c = C_1 x + C_2 x \ln(x)$

Rewriting the eqn

$$y'' - \frac{1}{x}y' + \frac{1}{x^2}y = \frac{2}{x}$$

$$W(x, x\ln x) = \begin{vmatrix} x & x\ln x \\ 1 & \ln x + 1 \end{vmatrix}$$

$$= x\ln x + x - x\ln x$$

$$= x$$

$$W_1 = \begin{vmatrix} 0 & x\ln x \\ \frac{2}{x} & \ln x + 1 \end{vmatrix}$$

$$= -2\ln x$$

$$W_2 = \begin{vmatrix} x & 0 \\ 1 & \frac{2}{x} \end{vmatrix}$$

$$= 2$$

$$u_1' = \frac{-2\ln x}{x} \quad , \quad u_2' = \frac{2}{x}$$

$$\Rightarrow u_1 = -(\ln x)^2 \qquad u_2' = 2\ln x$$

$$\therefore y = c_1 x + c_2 x\ln(x) - x(\ln x)^2 + 2x(\ln x)^2$$

$$= c_1 x + c_2 x\ln(x) + x(\ln x)^2$$

(2) $x^2y'' + xy' - y = 0$

Substituting $y = x^m$

$x^2 m(m-1)x^{m-2} + xmx^{m-1} - x^m = 0$

Auxiliary eqn is

$m(m-1) + m - 1 = 0$

$\Rightarrow m^2 - 1 = 0$

$\Rightarrow m = \pm 1$

$\therefore y_c = C_1 x + C_2 x^{-1}$

Rewriting the eqn

$$y'' + \frac{1}{x}y' - \frac{1}{x^2}y = \frac{\ln x}{x^2}$$

$$W(x, x^{-1}) = \begin{vmatrix} x & x^{-1} \\ 1 & -x^{-2} \end{vmatrix}$$

$$= -x^{-1} - x^{-1}$$

$$= -2x^{-1}$$

$$W_1 = \begin{vmatrix} 0 & x^{-1} \\ \frac{\ln x}{x^2} & -x^{-2} \end{vmatrix}$$

$$= -\frac{\ln x}{x^3}$$

$$W_2 = \begin{vmatrix} x & 0 \\ 1 & \frac{\ln x}{x^2} \end{vmatrix}$$

$$= \frac{\ln x}{x}$$

$$u_1' = \frac{+\ln(x)\,x}{2x^3} = \frac{\ln x}{2x^2}, \qquad u_2' = \frac{-\ln(x)\cdot(x)}{2x} = -\frac{\ln x}{2}$$

$$\Rightarrow u_1 = -\frac{1+\ln x}{2x}, \quad u_2 = -\frac{1}{2}(x\ln x - x)$$

$$y = c_1 x + c_2 x^{-1} - \frac{1}{2} - \frac{1}{2}\ln x - \frac{1}{2}\ln x + \frac{1}{2}$$

$$= c_1 x + c_2 x^{-1} - \ln(x)$$

Rewriting Cauchy-Euler Equation Into Linear Equation With Constant Coefficients

We make the substitution $x = e^t$, then solve the equation with constant coefficients in terms of t and resubstitute $t = \ln x$ in the final solution.

Example Solve

(1) $x^2 y'' + 9xy' - 20y = 0$

(2) $x^2 y'' - 3xy' + 13y = 4 + 3x$

Solution

(1). Substitute $x = e^t$ or $t = \ln x$

$$\frac{dy}{dx} = \frac{dy}{dt}\cdot\frac{dt}{dx} = \frac{1}{x}\frac{dy}{dt} = \frac{1}{e^t}\frac{dy}{dt}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dt}\right) = \frac{-1}{x^2}\frac{dy}{dt} + \frac{1}{x}\frac{d}{dx}\left(\frac{dy}{dt}\right)$$

$$\frac{d^2y}{dx^2} = -\frac{1}{x^2}\frac{dy}{dt} + \frac{1}{x}\frac{d^2y}{dt^2}\cdot\frac{1}{x} \qquad \text{since } \frac{dy}{dx} = \frac{1}{x}\frac{dy}{dt}$$

$$= -\frac{1}{x^2}\frac{dy}{dt} + \frac{1}{x^2}\frac{d^2y}{dt^2} = \frac{1}{e^{2t}}\left[\frac{d^2y}{dt^2} - \frac{dy}{dt}\right]$$

Substituting into the eqn

$$e^{2t}\left[\frac{1}{e^{2t}}\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right)\right] + 9e^t\frac{1}{e^t}\frac{dy}{dt} - 20y = 0$$

$$\Rightarrow \frac{d^2y}{dt^2} - \frac{dy}{dt} + 9\frac{dy}{dt} - 20y = 0$$

$$\Rightarrow \frac{d^2y}{dt^2} + 8\frac{dy}{dt} - 20y = 0$$

The auxiliary eqn is

$$m^2 + 8m - 20 = 0$$

$$\Rightarrow (m+10)(m-2) = 0$$

$$\Rightarrow m = -10 \text{ or } m = 2$$

$$\therefore y = Ae^{-10t} + Be^{2t}$$

Substituting $x = e^t$ gives

$$y = Ax^{-10} + Bx^2$$

(2) $x^2y'' - 3xy' + 13y = 0$

Let $x = e^t \Rightarrow t = \ln x$

$$\frac{dy}{dx} = \frac{dy}{dt}\cdot\frac{dt}{dx}$$

$$= \frac{1}{e^t}\frac{dy}{dt}$$

$$\frac{d^2y}{dx^2} = -\frac{1}{e^{2t}}\frac{dy}{dt} + \frac{1}{e^{2t}}\frac{d^2y}{dt^2}$$

Now substituting

$$e^{2t}\left[-\frac{1}{e^{2t}}\frac{dy}{dt} + \frac{1}{e^{2t}}\frac{d^2y}{dt^2}\right] - 3e^t\left[\frac{1}{e^t}\frac{dy}{dt}\right] + 13y = 0$$

$$\Rightarrow \frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 13y = 0$$

Auxiliary equation

$$m^2 - 4m + 13 = 0$$

$$m = \frac{4 \pm \sqrt{16 - 4(13)}}{2}$$

$$= \frac{4 \pm 6i}{2} = 2 \pm 3i$$

$$y_c = c_1 e^{2t}\cos 3t + c_2 e^{2t}\sin 3t$$

resubstituting x

$$y_c = c_1 x^2 \cos(3\ln x) + c_2 x^2 \sin(3\ln x)$$

Rewriting the eqn in standard form

$$y'' - \frac{3}{x}y' + \frac{13}{x^2}y = \frac{4}{x^2} + \frac{3}{x}$$

$$W(x^2\cos(3\ln x),\ x^2\sin(3\ln x))$$

$$= \begin{vmatrix} x^2\cos(3\ln x) & x^2\sin(3\ln x) \\ 2x\cos(3\ln x) - 3x\sin(3\ln x) & 2x\sin(3\ln x) + 3x\cos(3\ln x) \end{vmatrix}$$

$$= 2x^3\cos(3\ln x)\sin(3\ln x) + 3x^3\cos^2(3\ln x) - 2x^3\cos(3\ln x)\sin(3\ln x) + 3x^3\sin^2(3\ln x)$$

$$= 3x^3$$

$$W_1 = \begin{vmatrix} 0 & x^2\sin(3\ln x) \\ \frac{4}{x^2} + \frac{3}{x} & 2x\sin(3\ln x) + 3x\cos(3\ln x) \end{vmatrix}$$

$$= -4\sin(3\ln x) - 3x\sin(3\ln x)$$

$$W_2 = \begin{vmatrix} x^2\cos(3\ln x) & 0 \\ 2x\cos(3\ln x) - 3x\sin(3\ln x) & \frac{4}{x^2} + \frac{3}{x} \end{vmatrix}$$

$$= 4\cos(3\ln x) + 3x\cos(3\ln x)$$

$$u_1' = -\frac{4}{3x^3}\sin(3\ln x) - \frac{\sin(3\ln x)}{x^2}$$

$$u_2' = \frac{4\cos(3\ln x)}{3x^3} + \frac{\cos(3\ln x)}{x^2}$$

Using substitution along with integration by parts, we get

$$u_1 = \frac{4}{13}\left[\frac{2\sin(3\ln x)}{3x^2} + \cos(3\ln x)\right] + \frac{1}{10x}\left[\sin(3\ln x) + 3\cos(3\ln x)\right]$$

$$u_2 = \frac{4}{13}\left[-\frac{2}{3}\frac{\cos(3\ln x)}{x^2} + \frac{\sin(3\ln x)}{x^2}\right] + \frac{1}{10}\left[-\frac{\cos(3\ln x)}{x} + \frac{3\sin(3\ln x)}{x}\right]$$

$$\therefore y_p = u_1 x^2\cos(3\ln x) + u_2 x^2 \sin(3\ln x)$$

$$= \frac{4}{13}\left[\frac{2}{3}\sin(3\ln x)\cos(3\ln x) + \cos^2(3\ln x)\right] + \frac{x}{10}\left[\sin(3\ln x)\cos(3\ln x) + 3\cos^2(3\ln x)\right] + \frac{4}{3}\left[-\frac{2}{3}\cos(3\ln x)\sin(3\ln x) + \sin^2(3\ln x)\right] + \frac{x}{10}\left[-\cos(3\ln x)\sin(3\ln x) + 3\sin^2(3\ln x)\right]$$

$$= \frac{4}{13} + \frac{3}{10}x$$

$$\text{so } y = c_1 x^2\cos(3\ln x) + c_2 x^2 \sin(3\ln x) + \frac{4}{13} + \frac{3}{10}x.$$